

Damath

Damath is an educational board game originating from the Philippines, blending the traditional Filipino game "dama" with mathematical concepts, hence its name. It was invented by Jesus L. Huenda, a public high school teacher in Sorsogon, who encountered difficulties in teaching math using traditional methods. Inspired by a student project called "Dama de Numero" submitted by Emilio Hina Jr. in 1975, Huenda overhauled the game and introduced it to his class, who enjoyed playing it until it is commonly played in all elementary and secondary schools in the Philippines. Huenda's innovative approach has been recognized nationally and was awarded in 1981 with presidential merit medal, and the game has equipped Filipino students with essential numeracy skills. Damath encourages strategic thinking and problem-solving through creative activities like storytelling, flowcharts, and simulations.

Game Setup

The board consists of 64 squares arranged in a pattern of alternating black and white, similar to a chessboard. The four fundamental mathematical operations (+, -, *, /) are positioned on the white squares. Each square is labeled using a notation indicating its column and row. For instance, the square at the top-left corner is located at column 0 and row 7, identified as (0, 7). The chips' colors for the players are either red or blue, each starting with 12 pieces.

The Damath board consists of 64 spaces, but only half of them, the white spaces, are usable for gameplay. These numbered spaces represent the valid positions where pieces can travel, and they are as follows: **0, 2, 4, 6, 9, 11, 13, 15, 16, 18, 20, 22, 25, 27, 29, 31, 32, 34, 36, 38, 41, 43, 45, 47, 48, 50, 52, 54, 57, 59, 61, 63**. Pieces can only move to these numbered spaces, which form the playable grid of the game.

The Damath board associates specific positions with mathematical operators that determine the operation applied during a capture move. Addition is associated with positions 6, 13, 18, 25, 38, 45, 50, and 57, while Subtraction corresponds to positions 4, 15, 16, 27, 36, 47, 48, and 59. Multiplication is linked to positions 0, 11, 20, 31, 32, 43, 52, and 63, whereas Division is represented by positions 2, 9, 22, 29, 34, 41, 54, and 61. These operators play a crucial role in the game's scoring mechanics, as they define how values are calculated during captures.

The Damath board consists of 64 spaces, where half of the spaces can either hold a piece or remain empty. Each piece has three attributes: Color, which can be either red or blue; Value, represented as an integer that can be positive or negative; and Dama, a boolean indicating whether the piece has been promoted to a Dama (true for promoted, false otherwise). These attributes define the state and movement capabilities of each piece during the game.

Since Damath is derived from the traditional game of Dama, it follows some of its basic mechanics but introduces unique twists to incorporate mathematical elements. The following are its standard mechanics:

Earning the "Dama" Chip: A dama chip is an upgraded piece that provides an advantage by moving through multiple squares. A player's chip becomes a dama when it reaches the opponent's squares 0, 2, 4, 6. Similarly, the opponent's chip becomes a dama upon reaching the player's squares 57, 59, 61, 63.

Concluding the Game:

The game concludes after 20 minutes or if a player cannot move any chips because they are "trapped." The final score is the sum of points from capturing opponent chips and the numbers on remaining chips on the board, with dama chips' scores doubled. The player with the highest final score wins.

Piece Movement

Moving the Chips: Players can only slide their chips diagonally forward into an empty adjacent white square. Backward slides are not allowed unless capturing an opponent's chip. Players take turns moving a chip, and passing a turn is not permitted. Each player has 60 seconds per turn to make a move. If the chip is "Dama", it can move diagonally long forward or backward to any empty square until it reaches an allied chip.

A piece has a valid move if it is the player's turn, there are unoccupied spaces in front of the piece for a standard move, or an enemy piece is directly ahead or behind it with a free space beyond the enemy piece in the same direction for capturing.

For **Dama pieces**, they can move diagonally, as far as possible, until a piece blocks the way. If a blocking piece is present, all spaces behind it are not valid moves, except if the blocking piece is an enemy piece. If two adjacent pieces are adjacent on that diagonal, the Dama piece can only be moved up until the blocking piece, even if it is an enemy piece.

A piece can only move if it is the player's turn, prioritizing capturing when an enemy piece is directly in front or behind with a free space beyond for capture. Standard moves require free spaces ahead, and pieces cannot move backward unless they are a Dama or capturing an enemy piece.

Capturing an Opponent's Chip and Scoring if Ordinary Chip:

If an ordinary chip is to capture an opponent's chip, a player must "jump" over it diagonally within a diagonally adjacent tile next to the piece that is about to be taken, landing on the empty white square adjacent to the captured chip. Captures can be made diagonally forward or backward, and players cannot deny a capture. Upon capturing a chip, the capturing player scores points based on the mathematical operation indicated on the white square where the chip lands. This operation is applied to the numbers on the capturing and captured chips.

Capturing an Opponent's Chip and Scoring if Dama Chip:

If the chip is "Dama", they capture opponent chips by jumping over them diagonally no matter how long the distance between the two chips is, as long as there is a vacant space behind the chip that is about to be taken. The score from the dama chip's capture is calculated by doubling the result of the mathematical operation indicated on the landing square. The score is also doubled if an ordinary chip captures a dama chip and if a dama chip captures another dama chip the score would be quadrupled.

Piece Capturing

One valid move for a piece is capturing an enemy piece. If any piece has a move that captures an enemy, a **mandatory capture** occurs. When this happens, all other valid moves for other pieces are removed, unless they are also mandatory captures. If a Dama piece has a mandatory capture, its moves are prioritized over non-Dama pieces, even if those non-Dama pieces have mandatory capture moves.

Piece Promotion

A **Dama piece** is a piece that reaches the farthest row on the enemy's side. Once a piece reaches the last row, it is promoted to a Dama piece, unlocking new abilities. A Dama piece can move diagonally in any direction, provided the moves are valid. However, a Dama cannot jump over (capture) two or more adjacent pieces. When a Dama piece captures an enemy piece, the score will be higher than when a normal piece performs the capture.

In Damath, movements are categorized into two types: normal movement and capture movement. **Normal movement** involves a piece moving to a free square on the board, represented by the "Move" key in the notation. The "source" object specifies the piece and its current position, while the "destination" object indicates the target position, which must be empty (null for the "piece" value). For example, a red piece with a value of -5 moving from position 2 to position 4 is written as:

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{"Move":{"source":{"position":2,"piece":["red",-5,false]},"destination":{"position":4,"piece":null}}}
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Capture movement, represented by the "Capture" key, involves jumping over an enemy piece and landing on a free square behind it. It includes an additional "middle" object specifying the captured piece, which must be of the opposite color of the "source" piece. The "score" key reflects the result of the capture, determined by the operator on the destination square. For instance, a red piece with a value of -5 at position 2 captures a blue piece with a value of 6 at position 11, landing at position 20, with a score of -30, written as:

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{"Capture":{"source":{"position":2,"piece":["red",-5,false]},"middle":{"position":11,"piece":["blue",6,false]},"destination":{"position":20,"piece":null},"score":-30}}
```

These notations ensure clear representation of both standard moves and captures during gameplay.

In Damath, a piece can only make a valid move if several conditions are met: It must be the player's turn, and there must be free spaces in front of them. For capturing moves, there must be an enemy piece in front or behind the piece, and a free space behind the enemy piece in the same direction. Dama pieces, which are promoted pieces, can move diagonally as far as possible, but their movement is blocked by other pieces. If a blocking piece is present, all spaces behind it are invalid unless the blocking piece is an enemy. If two adjacent pieces are positioned diagonally, the Dama piece can only move up to the blocking piece, even if it is an enemy. The priorities for valid movement are: ensuring it is the player's turn, checking for an enemy piece in the direction of movement, and confirming free spaces in front. Additionally, a piece cannot move backward unless it is a Dama piece or the move involves capturing an enemy piece.

In Damath, a Dama piece is a piece that reaches the farthest row on the enemy's side of the board. Upon reaching this row, the piece is promoted to a Dama, unlocking new movement abilities. A Dama piece can move diagonally in any direction, as long as the moves are valid. However, a Dama cannot jump over (capture) two or more adjacent pieces. When a Dama captures an enemy piece, the score associated with the capture is higher compared to a normal piece performing the same action, reflecting its enhanced capabilities.

In Damath, **piece capturing** is a valid move where a piece jumps over an enemy piece and lands on a free square behind it. If a piece has a move that captures an enemy, a **mandatory capture** occurs, meaning the capture must be performed. Once a mandatory capture is made, all other valid moves for other pieces are removed, unless they also involve mandatory captures. If a Dama piece has a mandatory capture, its move is prioritized over those of non-Dama pieces, even if the non-Dama pieces also have mandatory capture moves, ensuring that Dama pieces take precedence in capturing scenarios.

In Damath, the score for a piece capture is calculated using the formula: $\text{score} = \text{pieceValue} \{\text{operator}\} \text{enemyPieceValue}$, where pieceValue is the value of the capturing piece, operator is the mapped mathematical operator based on the destination square, and enemyPieceValue is the value of the captured enemy piece. For Dama pieces, the calculated score is doubled, and if an ordinary piece captures a Dama piece, the score is also doubled. When a Dama piece captures another Dama piece, the score is quadrupled. These scoring rules emphasize the higher value of Dama pieces in gameplay and reward strategic captures accordingly.